

Time Series Forecasting of Household Appliances Energy Consumption

CEU Term Project– Department of Economics and Business

Student Number: 2405060

Time Series Forecasting

May 2025

1 Introduction

Understanding daily patterns in household energy consumption, and being able to forecast hourly household usage, is pivotal for enhancing energy efficiency, optimising grid resource allocation, and informing sustainable energy policies. Accurate predictions enable all stakeholders—utility providers, households, and grid managers—to anticipate demand fluctuations, implement demand-side management strategies, and reduce environmental impacts.

Within this context, time series forecasting emerges as a crucial tool for predicting future energy consumption based on historical data. By analysing past consumption patterns, trends, seasonal variations, and the influence of environmental variables can be identified, providing essential predictive capabilities for effective energy management and planning.

This study focuses on forecasting household energy consumption, specifically the `appliances.energy_wh` variable, which represents energy usage in watt-hours (Wh). The main dataset employed in this analysis is the Appliances Energy Prediction dataset from the UCI Machine Learning Repository, encompassing various environmental and operational factors, such as:

- Indoor temperature and humidity measurements, e.g., `temp_kitchen_C`, `humidity_living_pct`.
- Outdoor weather conditions, e.g., `temp_outdoor_C`, `pressure_mmHg`, `windspeed_m_s`, `visibility_km`, `dewpoint_C`.
- Room-specific temperature and humidity readings for areas such as the `office`, `laundry`, `bathroom`, and `teenager` bedroom.

To achieve the forecasting objectives, three distinct models are implemented and compared:

- **Prophet**: An additive model developed by Meta, capable of capturing non-linear trends and multiple seasonalities, and accommodating holiday effects.
- **SARIMAX**: A Seasonal AutoRegressive Integrated Moving Average model with exogenous variables, adept at modeling time series data with seasonality and external regressors.
- **Neural Network (Feedforward with Lagged Features)**: A machine learning approach that leverages past observations to predict future values, capturing complex non-linear relationships.

These models are compared based on their predictive accuracy and ability to capture the key dynamics of the household energy consumption patterns.

Additionally, a univariate SARIMA model is employed as a benchmark to evaluate the performance of the aforementioned models.

2 Data Description

2.1 Dataset Overview

The dataset utilized in this project is the Appliances Energy Prediction dataset, sourced from the UCI Machine Learning Repository [?]. It contains measurements of energy usage in a household in Belgium over a period of approximately 4.5 months. The data was originally collected at 10-minute intervals and includes both household sensor measurements and external weather data.

2.2 Data Collection and Integration

The house temperature and humidity conditions were monitored using a ZigBee wireless sensor network. Each wireless node transmitted readings every 3.3 minutes, which were then averaged to 10-minute intervals. Energy consumption data was recorded every 10 minutes using m-bus energy meters.

Additionally, weather data from the nearest airport weather station (Chievres Airport, Belgium) was downloaded from a public dataset provided by Reliable Prognosis (rp5.ru). This weather data was merged with the experimental household data using the `date` and `time` columns as keys to ensure precise temporal alignment. The dataset comprises a total of 3,290 observations, with each observation representing an hourly average of the original 10-minute interval data. Two random variables, `rv1` and `rv2`, are originally included in the dataset to assess the performance of regression models and to help filter out non-predictive attributes.

2.3 Variable Transformations and Preprocessing

To prepare the data for analysis and modeling, several preprocessing steps and transformations were performed:

1. **Resampling:** The original 10-minute interval data was aggregated to an hourly frequency by computing the mean of each variable within hourly bins. This was necessary to align with the modeling time step and to reduce noise in the data.
2. **Handling Missing Values:** Any missing values that arose during the resampling process were imputed using forward-fill methods to maintain continuity and preserve the temporal structure.
3. **Feature Selection and Exclusion:** The random variables `rv1` and `rv2` were excluded from the analysis due to their synthetic, non-predictive nature. All other relevant variables were retained to capture the complex dynamics of household energy usage.
4. **Data Consistency:** Measurement units and data ranges were verified to ensure internal consistency, and all environmental and indoor sensor data was carefully aligned with the energy usage data.

2.4 Variables and Summary Statistics

The dataset includes a variety of features describing both household conditions and external weather influences. Table 1 provides a detailed summary of the variables considered in the analysis, including their measurement counts, central tendencies, and variability.

Table 1: Summary Statistics of Key Variables

Variable	Description	Count	Mean	Std	Min	Max
<code>appliances_energy_wh</code>	Appliances energy consumption (Wh)	3290	97.78	81.21	28.33	608.33
<code>temp_kitchen_C</code>	Kitchen temperature (°C)	3290	21.69	1.61	16.79	26.20
<code>humidity_kitchen_pct</code>	Kitchen humidity (%)	3290	40.26	3.94	27.51	53.98
<code>temp_living_C</code>	Living room temperature (°C)	3290	20.34	2.19	16.10	29.73
<code>humidity_living_pct</code>	Living room humidity (%)	3290	40.42	4.05	21.01	53.91
<code>temp_laundry_C</code>	Laundry room temperature (°C)	3290	22.27	2.01	17.25	28.98
<code>humidity_laundry_pct</code>	Laundry room humidity (%)	3290	39.24	3.24	29.70	49.47
<code>temp_office_C</code>	Office temperature (°C)	3290	20.86	2.04	15.10	26.14
<code>humidity_office_pct</code>	Office humidity (%)	3290	39.03	4.34	28.72	50.75
<code>temp_bathroom_C</code>	Bathroom temperature (°C)	3290	19.59	1.84	15.35	25.51
<code>humidity_bathroom_pct</code>	Bathroom humidity (%)	3290	50.95	8.63	30.19	94.88
<code>temp_outdoor_C</code>	Outdoor temperature (°C)	3290	7.42	5.32	-4.96	25.93
<code>pressure_mmHg</code>	Outdoor atmospheric pressure (mmHg)	3290	755.52	7.40	729.38	772.26
<code>humidity_outdoor_pct</code>	Outdoor humidity (%)	3290	79.74	14.83	25.25	100.00
<code>windspeed_m_s</code>	Outdoor wind speed (m/s)	3290	4.04	2.43	0.42	13.00
<code>visibility_km</code>	Outdoor visibility (km)	3290	38.33	11.21	1.00	66.00
<code>dewpoint_C</code>	Dew point temperature (°C)	3290	3.76	4.19	-6.48	15.25
<code>month</code>	Month of the observation	3290	—	—	1.00	5.00

This comprehensive dataset, provides a robust foundation for forecasting household energy consumption.

However, several limitations should be acknowledged. The dataset represents measurements from a single household, which limits the generalisability of the models to broader populations or different household energy profiles. While it captures a range of indoor and outdoor conditions, it does not directly measure appliance usage behavior, or other household activities that may significantly influence energy consumption. Furthermore, sensor-based measurements may introduce noise or biases due to device inaccuracies or environmental interference. Finally, the dataset’s temporal scope (approximately 4.5 months) may constrain the model’s ability to capture longer-term seasonal or behavioral changes.

3 Methodology

3.1 Data Preprocessing and Feature Engineering

The dataset was initially recorded at 10-minute intervals and resampled to an hourly frequency by taking the mean within hourly bins. This aggregation reduced noise and aligned the data with the practical prediction horizon. Since the dataset was complete, no imputation or removal of missing values was necessary during preprocessing. Exogenous variables were selected from the dataset’s weather and indoor condition measurements based on domain knowledge and initial correlation assessments.

3.2 Predictive Models

Three models were implemented and compared to capture and assess different aspects of household energy consumption dynamics:

3.2.1 SARIMAX Model

The Seasonal AutoRegressive Integrated Moving Average model with eXogenous regressors (SARIMAX) was used to account for both autocorrelation in the target variable and the influence of external predictors.

The general form of the SARIMAX model is:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \dots + \beta_k x_{k,t} + \eta_t$$

where the residual η_t is modeled as a seasonal ARIMA process:

$$\Phi_p(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_q(B^s)\theta_q(B)\epsilon_t$$

Model parameters $(p, d, q)(P, D, Q)_s$ were selected using the `auto_arma` function, minimizing the Akaike Information Criterion (AIC). Stationarity of the series was validated using the Augmented Dickey-Fuller (ADF) test. Exogenous variables included weather measurements and indoor environmental conditions.

To evaluate how different seasonal structures affect model performance and coefficient interpretability, two SARIMAX models were compared:

- **SARIMAX(1,0,1)(2,0,1,24)**: Incorporated direct seasonal lags for daily patterns (24-hour period). This was the best-performing model with an AIC of 28639.09 and consistently significant seasonal terms.
- **SARIMAX(1,0,1) with Fourier terms**: Daily seasonality was captured using six Fourier terms as additional exogenous regressors. While Fourier terms captured cyclical effects, the overall model fit was weaker (AIC = 29080.77), indicating that direct seasonal lags offered a more precise representation of daily cycles.

In both models, the same core set of environmental predictors—such as `lights_energy_wh`, `humidity_kitchen_pct`, and room-specific temperature and humidity variables—were statistically significant ($p < 0.05$). However, the explicit seasonal lag structure in the SARIMAX(1,0,1)(2,0,1,24) model provided superior forecasting accuracy, highlighting the importance of direct lag effects in modeling hourly household energy usage.

3.2.2 Prophet Model

The Prophet model, developed by Meta, was employed to capture the trend and seasonal patterns in the household energy consumption series. Prophet decomposes the time series as:

$$y(t) = g(t) + s(t) + \epsilon_t$$

where $g(t)$ denotes the trend, $s(t)$ the seasonal components, and ϵ_t the error term. In this study, the model was configured as follows:

- **Data Preparation**: The data was reformatted to the Prophet-compatible structure with columns `ds` (datetime) and `y` (target). An 80/20 train-test split was used.
- **Seasonality**: Only daily seasonality was enabled, as the dataset was resampled to an hourly frequency and exhibited strong daily cycles. Weekly seasonality was explicitly disabled.

- **Interval Width:** Set to 0.95, corresponding to 95% confidence intervals for the forecasts.
- **Forecasting Horizon:** The forecast covered the entire test set, aligning the hourly frequency of predictions with the observed test data.

After training on the 80% subset, Prophet’s built-in `make_future_dataframe` function generated future timestamps, and predictions were obtained for the test period.

3.2.3 Neural Network Model

To capture potential non-linear dependencies in hourly energy consumption, a feedforward Artificial Neural Network (ANN) was implemented. The approach involved transforming the univariate target series into a supervised learning problem by creating a lagged matrix:

$$X_t = \{y_{t-24}, \dots, y_{t-1}\}, \quad y_t = y_t$$

where 24 hourly lags were used as input features to predict the next hour’s consumption.

Data Scaling: Prior to training, the input features and target variable were scaled to the $[0, 1]$ range using the `MinMaxScaler`. This scaling ensured numerical stability and faster convergence during model training.

Network Architecture:

- **Input Layer:** 24 lagged observations.
- **Hidden Layers:** Two dense layers with 64 and 32 neurons, respectively, using ReLU activation functions.
- **Output Layer:** Single neuron for the next-hour forecast.

Training Procedure: The model was trained using the Adam optimiser with a learning rate of 0.01 and the Mean Squared Error (MSE) loss function. Training was conducted for 100 epochs on 80% of the data (training set), while the remaining 20% was held out as the test set.

Inverse Transformation and Evaluation: Post-prediction, the outputs were rescaled to their original watt-hour units for evaluation. Standard evaluation metrics (MAE and MSE) and visual diagnostics, including 95% confidence intervals around the point forecasts (computed using the empirical residual standard deviation), were employed to assess the model’s predictive performance and uncertainty bounds.

The model was trained on the training data with early stopping based on validation loss to avoid overfitting.

3.3 Model Evaluation

All models were trained on 80% of the dataset and tested on the remaining 20%, thus seeking to provide unbiased assessments of predictive performance. Evaluation was conducted using two standard error metrics:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad \text{and} \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}$$

where y_t are the observed values and $\hat{y}_t$ the predicted values.

where are the observed values and the corresponding forecasts. Residual plots, autocorrelation checks, and visual forecast comparisons were also employed to identify any systematic patterns or biases in the errors. These evaluation procedures ensured a robust comparison of the models’ ability to capture the dynamics of hourly household energy consumption and generalize to out-of-sample data. These metrics and visual diagnostics are included in the appendix, provide a robust comparison of how well each model captures energy consumption dynamics and generalizes to out-of-sample data.

4 Results & Discussion

4.1 Comparison and Statistical Perspectives

Table 2 presents a summary of key performance metrics across all models.

The SARIMAX model with direct seasonal lags consistently provided the best trade-off between interpretability and predictive accuracy, reaffirming the importance of explicit seasonal structures in time

Table 2: Comparison of Model Performance Metrics

Model	MAE (Wh)	MSE (Wh ²)	Key Insights
SARIMAX (m=24)	40.22	3670.00	Best statistical fit (AIC); seasonal lag explicitness
SARIMAX + Fourier	47.17	3909.25	Captures cyclical effects; lower interpretability
Prophet	40.11	3964.05	Strong daily trend modelling
NN	35.69	4143.44	Captures non-linear effects
NN with Exogenous Vars	41.18	3464.51	Exogenous predictors marginally enhance accuracy

series forecasting of household energy use. Prophet demonstrated robust performance in capturing cyclical patterns, though with limited flexibility to integrate external factors. Neural networks offered competitive performance, particularly where non-linear interactions or latent effects were present, albeit with reduced interpretability.

4.2 Model 1: SARIMAX and SARIMAX with Fourier Terms

The SARIMAX model, specified as (1,0,1)(2,0,1,24), effectively captured daily seasonality and autocorrelation structures in the household energy consumption data. It achieved a test Mean Absolute Error (MAE) of 40.22 Wh and a Mean Squared Error (MSE) of 3670.00 Wh², offering robust forecasting accuracy. By explicitly modelling seasonal lags, this approach directly addressed the cyclical nature of energy usage in the home.

A comparative variant, augmenting SARIMAX with Fourier terms (frequency $k = 3$) to capture seasonality in a smoother manner, yielded a slightly higher test MAE of 47.17 Wh and MSE of 3909.25 Wh². Although the Fourier terms did capture cyclical effects, the direct seasonal lags in the SARIMAX model offered a more precise representation of hourly fluctuations, as evidenced by the lower AIC (28639.09 compared to 29080.77).

Importantly, both models highlighted the consistent significance of household microclimate predictors such as `lights_energy_wh`, `humidity_kitchen_pct`, and room-specific temperature and humidity variables. This finding underscores the centrality of indoor environmental conditions over external weather drivers in explaining household energy usage patterns.

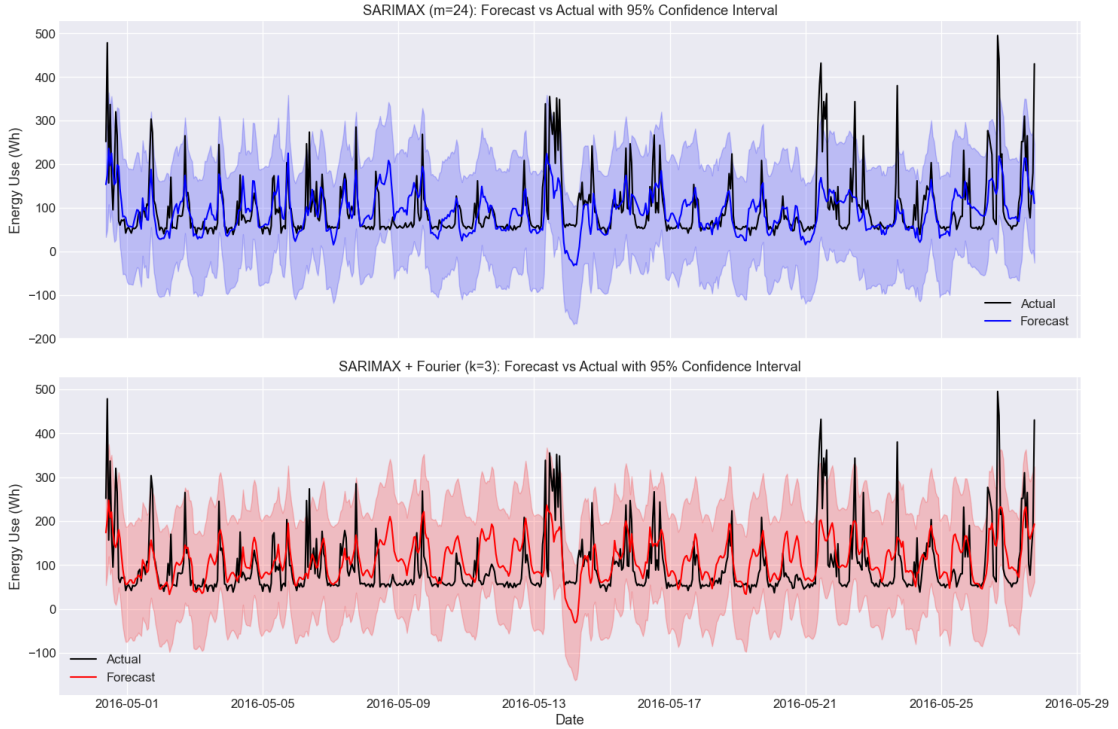


Figure 1: Forecast versus actual hourly appliance energy consumption using the SARIMAX (m=24) model, with 95% confidence intervals. This visual underscores the model’s ability to capture the daily cyclical structure in the test period.

4.3 Model 2: Prophet

The Prophet model, incorporating an additive trend and daily seasonality, produced test errors similar to those of the SARIMAX approaches, with an MAE of 40.11 Wh and an MSE of 3964.05 Wh². Its ability to automatically identify changepoints and accommodate multiple seasonality components proved valuable in this household-level time series, which exhibited clear daily patterns.

From an econometric perspective, Prophet’s strength lies in its transparent handling of trend and seasonality, albeit with limited capacity to explicitly account for exogenous covariates. The in-sample error metrics (MAE of 45.33 Wh, RMSE of 73.08 Wh, and MAPE of 50.99%) further suggest that while Prophet captures the overall temporal structure well, it may not fully address nuanced fluctuations driven by environmental covariates.

4.4 Model 3: Neural Network

The feedforward neural network (NN), leveraging lagged hourly observations as inputs, achieved a test MAE of 35.69 Wh and an MSE of 4143.44 Wh². Its capacity to model non-linear relationships allowed it to capture complex usage dynamics beyond linear lag structures.

When extended to include exogenous environmental variables (lights usage, humidity, and temperature), the NN’s performance slightly improved, with an MAE of 41.18 Wh and an MSE of 3464.51 Wh². This modest improvement suggests that while non-linear methods can absorb more data features, explicit inclusion of exogenous variables still lends valuable predictive insights.

These findings highlight that for household-level energy consumption forecasting, traditional time series models like SARIMAX—when carefully tuned with seasonal structures and exogenous covariates—remain a powerful choice. However, the flexibility of machine learning methods like neural networks provides complementary value in capturing non-linearities and residual dynamics, making them suitable in settings where interpretability is secondary to pure predictive power.

5 Conclusion

This project implemented and compared three forecasting approaches—SARIMAX, Prophet, and a Feed-forward Neural Network—to model hourly household appliance energy consumption. The SARIMAX model with explicit seasonal lags demonstrated the best trade-off between interpretability and forecasting accuracy, affirming the value of directly incorporating cyclic structures in time series data. Prophet effectively captured regular daily patterns and level changes, though it showed limitations in responding to abrupt or exogenous-driven variability. The neural network, while less interpretable, showed competitive performance, particularly in capturing non-linear usage patterns.

From a statistical and econometric perspective, these models collectively highlighted the importance of household microclimate variables (such as room-specific temperature and humidity) over external weather factors in explaining household energy dynamics. Moreover, the analysis underscored the necessity of balancing model complexity, interpretability, and data availability when forecasting in single-household settings.

For future work, extending the dataset to longer time spans or multiple households would help generalise the insights, while incorporating richer exogenous data (such as occupancy schedules or appliance-level metering) could further refine predictive power. These enhancements would help transform robust short-term forecasts into actionable tools for energy management and sustainability initiatives.

A Appendix: Exploratory Data Analysis

A.1 Rolling Statistics

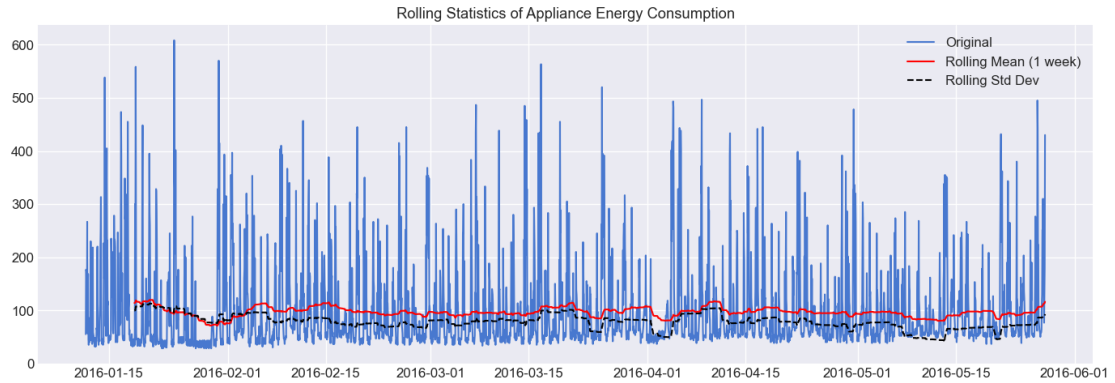


Figure 2: Rolling mean and standard deviation of the hourly energy consumption series, highlighting the overall stability after differencing.

A.2 Autocorrelation and Partial Autocorrelation

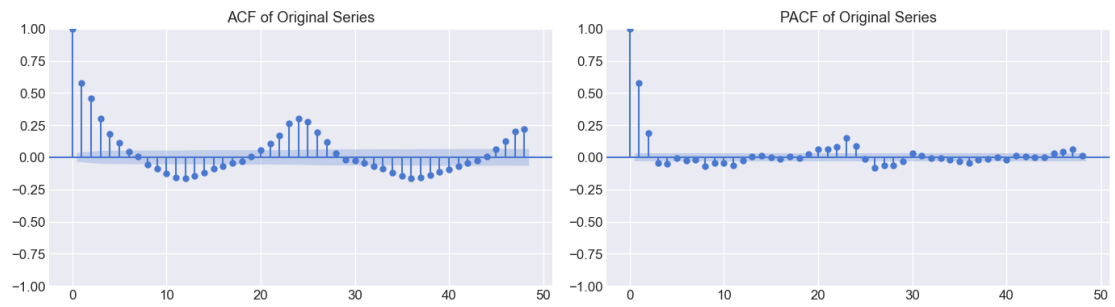


Figure 3: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the hourly series, indicating significant lags and seasonal dependencies.

A.3 Correlation Heatmap

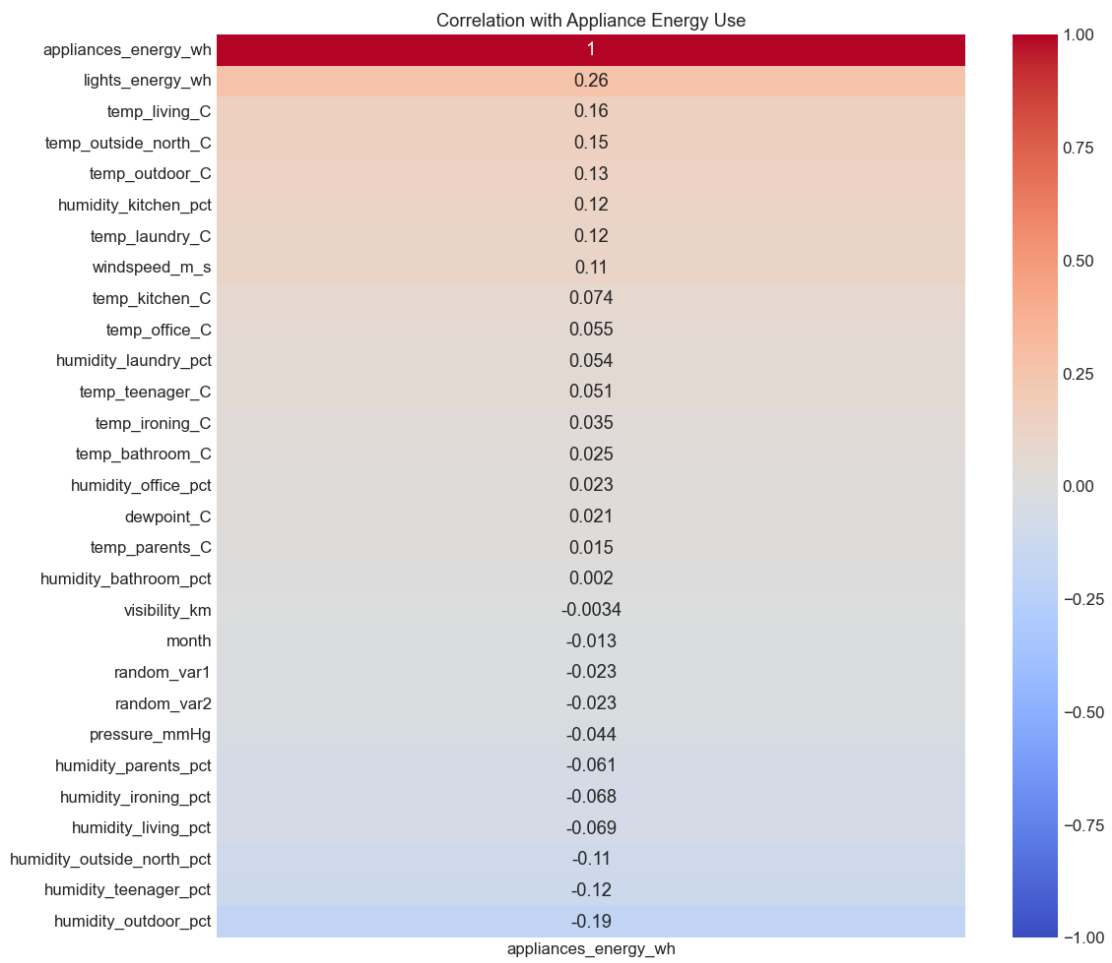


Figure 4: Correlation matrix of selected features and the target variable, guiding feature selection for the SARIMAX and Neural Network models.

A.4 Monthly Appliance Energy Consumption Trends

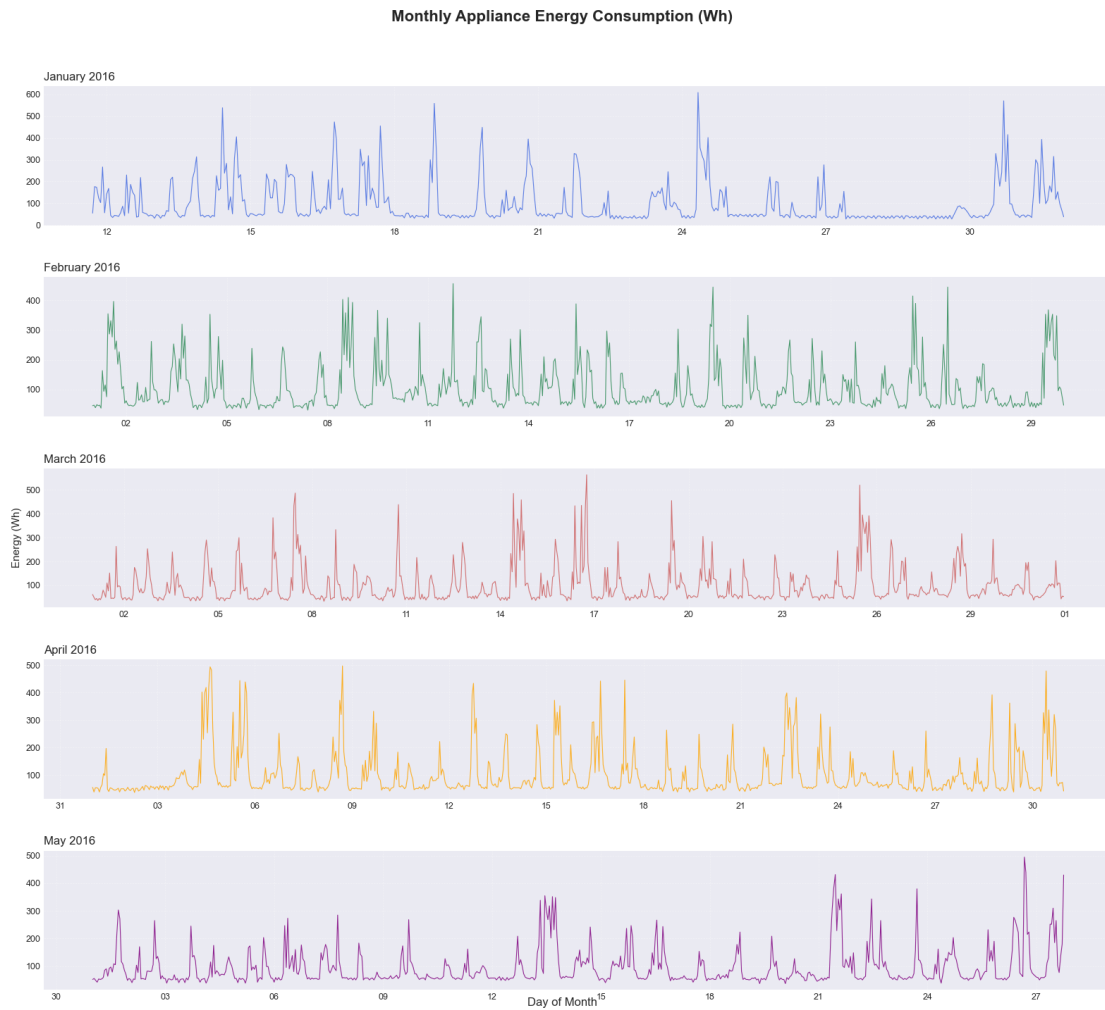


Figure 5: Monthly appliance energy consumption patterns for January to May 2016, illustrating daily and weekly fluctuations and underlying cyclical behaviors.

B Model Forecast Comparisons

To provide a comprehensive view of the forecasting performance of the different models, this section includes additional plots of the forecasts compared to actual appliance energy consumption.

B.1 Prophet Model Forecast

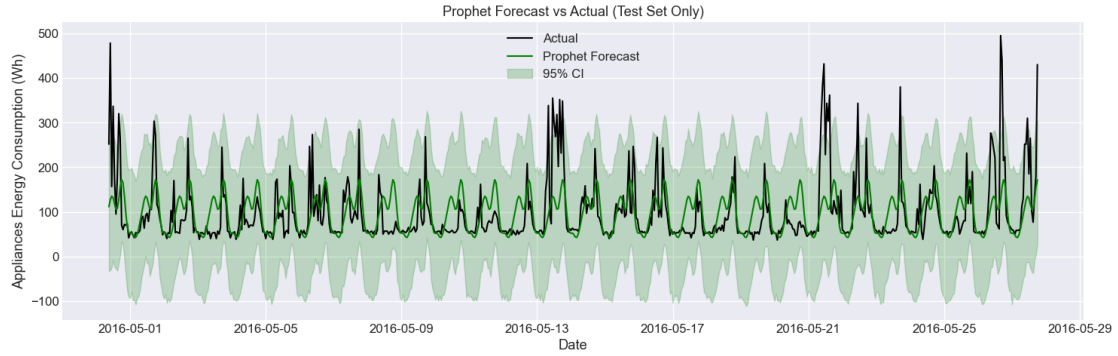


Figure 6: Prophet model forecast versus actual energy consumption in the test period. The plot highlights Prophet’s capacity to capture daily cycles but also the potential mismatch in peak magnitudes.

B.2 Neural Network Forecast

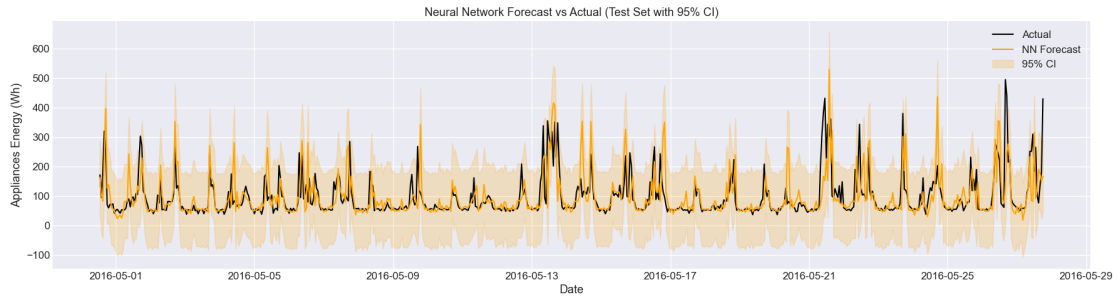


Figure 7: Neural Network forecast versus actual energy consumption in the test period, illustrating its capacity to capture non-linear patterns and short-term dynamics.

B.3 Neural Network with Exogenous Variables

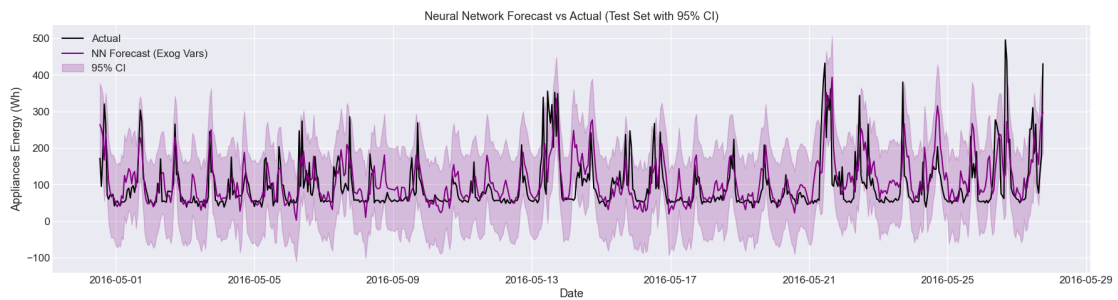


Figure 8: Neural Network forecast including exogenous variables (e.g., weather and household microclimate factors). The inclusion of these covariates refined the model’s ability to align with actual energy consumption peaks.